



BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-2027
COURSE HANDOUT (Part II)

Date: 04.08.2026

In addition to Part I (General Handout for all courses appended to the time table) portion:

Course Number : MATH F432
Course Title : Applied Statistical Methods
Instructor-In charge : SUMANTA PASARI
Instructor : Niladri Chatterjee and Sumanta Pasari

1. Course Description:

Applied Statistics is an exciting sub-area of Statistics. It has application in almost all science disciplines which deal with data and uncertainty. This course builds up fundamental concepts of various key statistical methods necessary to analyze/interpret a variety of practical business/engineering problems. This course emphasizes the role of statistics in one's own field of study by making sense of data, developing self-ability to apply appropriate statistical methods, performing experimental designs and above all, realizing the limitations/inherent assumptions in a statistical test to avoid over interpretation or misinterpretation. The thrust areas covered in this course include probability distributions, statistical inference, analysis of variance (ANOVA), regression and correlation, discriminant analysis, factor and cluster analysis, time series analysis and forecasting, nonparametric methods and statistical quality control. The theoretical learning will be complemented with various case studies and hands-on training in excel/R.

2. Scope and objective of the course:

This 4th level course is designed with a two-fold purpose. First, it will provide an exposure to various theoretical univariate/multivariate methods and their practical applications. Second, this course will improve methodological/analytical maturity to attempt a variety of problems using MS-excel and R.

3. Text Book:

T1: David R Anderson, Dennis J Sweeney, Thomas A Williams, Jeffrey D. Camm and James J. Cochran, Statistics for Business and Economics, 13th Edition, Cengage Learning, 2019.

4. Reference Books:

1. Deepak Chawla and Neena Sondhi, Research Methodology, Vikas, 2012.
2. Richard Johnson and Dean W Wichern, Applied Multivariate Statistical Analysis, Pearson, 2007.

5. Course Plan:

Module	Topics to be covered	Reference (Chapter number)	Learning Outcomes
Module 1	L1-2: Revision of basic probability concepts, random variables, probability distributions and moments.	Chapter 4 to Chapter 6 (T1)	The purpose is to recapitulate the introductory probability concepts.
	L3-4: Introduction, Review of sampling, Selecting a sample, Sampling from a finite and infinite population,	Chapter 7 (T1)	It enhances the understanding of different sampling procedures, sampling



BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
Pilani Campus
AUGS/ AGSR Division

	Point Estimation, Sampling distribution of sample mean and properties of point estimators, Other sampling methods		distribution and inferential procedures.
	L5-6: Interval estimation, Population mean when σ is known, Population mean σ is unknown, Determining the sample size; Introduction to hypothesis testing, Types of errors, Inferences about single population mean.	Chapter 8 (T1) Chapter 9 (T1)	Students learn the concepts of hypotheses and its errors followed by decision scenarios.
Module 2	L7-8: Inferences about difference of two means, Inferences about population variances, Interval estimation. Testing the equality of population proportions, test of independence, and goodness of fit test.	Chapter 10 (T1) Chapter 11 (T1) Chapter 12 (T1)	Students will gain knowledge on importance of variance, inference about population variances, chi-square distributions and its types.
	L9-10: Introduction to experimental design and Analysis of Variance (ANOVA) (Completely randomized design, Multiple comparison procedures, Randomized block design)	Chapter 13 (13.1,13.2,13.3,13.4) (T1)	Students learn concepts of experimental design and ANOVA.
Module 3	L11-14: Concepts of non-parametric tests: Sign test, Wilcoxon signed rank test, Mann Whitney Wilcoxon test, Kruskal walls test, K-S two sample test	Chapter 18 (18.1-18.4) (T1), Class Notes	Students learn various concepts of non-parametric methods in parallel to parametric procedures.
Module 4	L15-18: Forecasting, Components of a time series, Smoothing methods, Trend projections, Trend and seasonal components,	Chapter 17 (17.1 – 17.6) (T1)	Students learn the concept of time series and associated time series forecasting.



BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
Pilani Campus
AUGS/ AGSR Division

	Regression analysis, Qualitative approaches.		
Module 5	L 19-20: Introduction, Control charts for variables, Control charts attributes, Modified control charts.	Chapter 19 (19.1, 19.2) (T1), Class Notes	Students gain understanding of various statistical quality control measures.
Module 6	L21-23: Simple linear regression model, Least squares method, Coefficient of determination, Model assumptions, Test for significance, Using the estimated regression equation for estimation and prediction, Residual analysis: Validating model assumptions, outliers and influential observations.	Chapter 14 (T1)	Students will learn basic regression model.
	L24-25: Multiple regression model, Least squares method, Multiple coefficient of determination, Model assumptions, Testing for significance, Multicollinearity. Regression equation for estimation and prediction, residual analysis, Discuss case studies.	Chapter 15 (15.1-15.6, 15.8) (T1)	It helps in understanding more than two variables in regression analysis and also gives insight on the concept of multicollinearity.
	L 26-27: Categorical independent variable, Logistic regression.	Chapter 15 (15.7,15.9) (T1)	It gives exposure to distinguish between categorical independent and categorical dependent regression analysis.
Module 7	L28-32: Principal Component Analysis; Factor analysis and its uses, Conditions for a factor analysis, Illustration of factor analysis, Applications	Chapter 16 (R1), Class Notes	It helps students in understanding data reduction methods.



BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
Pilani Campus
AUGS/ AGSR Division

	of factor analysis in other multivariate technique.		
Module 8	L33-35: Hotelling T^2 and Mahalanobis D^2 discriminant analysis, Objectives and its uses, Illustration of discriminant analysis, Assessing classification accuracy.	Chapter 17 (R1), Class Notes	It helps in assessing the classification accuracy of model.
Module 9	L36-40: Cluster analysis- a classification technique; Statistics associated with cluster analysis; An illustration of the technique, Key concepts in cluster analysis; Process of clustering; Establishing cluster algorithms; Discuss case studies	Chapter 18 (R1), Class Notes	It helps in understanding hierarchical and non-hierarchical cluster analysis.
Module 10	Group presentations		

6. Evaluation Scheme:

Component	Durations	Weightage	Date	Nature
Mid-Semester Test	90 min	25%	October 07, 2026	Open/Closed Book*
Assignments (two)	1-2 weeks	20%	Will be announced later	Open Book
Quizzes (best three out of four)	20 min	15%	Will be announced later	Closed Book
Comprehensive	180 min	40%	December 08, 2026	Open Book*

* For open book in midsem/comprehensive exam, you can bring only textbook and hand-written class notes.

7. Consultation hours: To be announced in class.

8. Notices: All notices in relation to the above course will be displayed on NALANDA.

9. Make up policy: Make up for mid-sem/comprehensive examination will be granted only in genuine cases. **There is no make-up for quiz components.** Students should follow proper guideline for applying make ups.

Instructor in Charge
MATH F432